

Derivate di distribuzioni

Def. ($n=1$, $\Omega \subseteq \mathbb{R}$)

Data $T \in \mathcal{D}'(\Omega)$ definiamo $T' \in \mathcal{D}'(\Omega)$ come

$$\boxed{\langle T', \varphi \rangle := - \langle T, \varphi' \rangle} \quad \forall \varphi \in \mathcal{D}(\Omega)$$

Osservazioni.

• $\varphi \in \mathcal{C}_0^\infty(\Omega) \Rightarrow \varphi' \in \mathcal{C}_0^\infty(\Omega) \Rightarrow \langle T, \varphi' \rangle$ ha senso

• 1) lineare

$$\langle T', \alpha_1 \varphi_1 + \alpha_2 \varphi_2 \rangle \stackrel{?}{=} \alpha_1 \langle T', \varphi_1 \rangle + \alpha_2 \langle T', \varphi_2 \rangle$$

$$\stackrel{||}{=} - \langle T, (\alpha_1 \varphi_1 + \alpha_2 \varphi_2)' \rangle \quad \stackrel{||}{=} - \alpha_1 \langle T, \varphi_1' \rangle - \alpha_2 \langle T, \varphi_2' \rangle$$

$$\stackrel{||}{=} - \langle T, \alpha_1 \varphi_1' + \alpha_2 \varphi_2' \rangle$$

$$\stackrel{||}{=} - \alpha_1 \langle T, \varphi_1' \rangle - \alpha_2 \langle T, \varphi_2' \rangle$$

$$2) \varphi_n \rightarrow 0 \text{ in } \mathcal{D}(\Omega) \stackrel{?}{\Rightarrow} \langle T', \varphi_n \rangle \rightarrow 0$$

$$\Downarrow \\ \varphi_n' \rightarrow 0 \text{ in } \mathcal{D}(\Omega)$$

$$\stackrel{||}{=} - \langle T, \varphi_n' \rangle$$

• Perché definire T' così ??

Se considero $T = T_u$ con $u \in \mathcal{C}^1(\Omega) \subseteq L^1_{loc}(\Omega)$

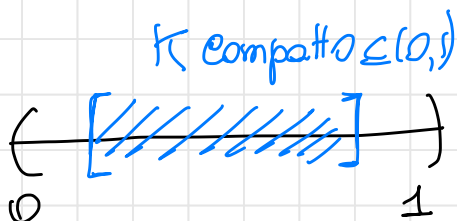
Ho anche considerare $T_{u'}$ $u' \in \mathcal{C}^0(\Omega) \subseteq L^1_{loc}(\Omega)$

$$T_{u'} \stackrel{?}{=} (T_u)' \quad \text{in } \mathcal{D}'(\Omega)$$

$$\bullet \langle T_{u'}, \varphi \rangle = \int_{\Omega} u' \varphi \quad \Omega = (0,1)$$

$$\bullet \langle (T_u)', \varphi \rangle = - \langle T_u, \varphi' \rangle$$

$$= - \int_{\Omega} u \varphi'$$



~~$$+ u \varphi \Big|_0^1$$~~

perché
 $\varphi \in \mathcal{C}_0^\infty(\Omega)$

Def. $\Omega \subseteq \mathbb{R}^n$, $T \in \mathcal{D}'(\Omega)$

$$\langle \mathcal{D}^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, \mathcal{D}^\alpha \varphi \rangle$$

$$\alpha = (\alpha_1, \dots, \alpha_n)$$

Esempi

1) $T = T_u$ con $u \in \mathcal{D}^1(\Omega)$ ($u, u' \in L^1_{loc}(\Omega)$)
 $(T_u)' = T_{u'}$

2) $T = T_u$ con $u(x) = |x|$ con $\Omega = (-1, 1)$

$$\langle (T_{|x|})', \varphi \rangle = - \langle T_{|x|}, \varphi' \rangle =$$

$$= - \int_{-1}^1 |x| \varphi' = - \int_0^1 x \varphi' + \int_{-1}^0 x \varphi'$$

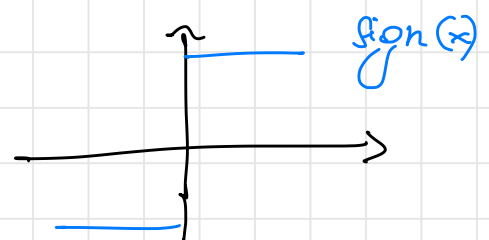
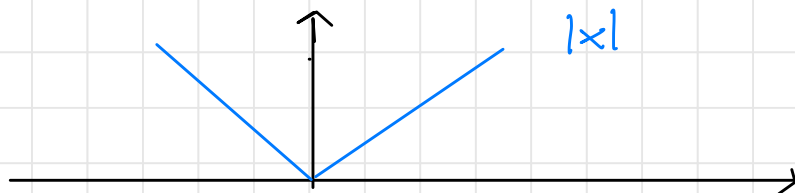
$$= \int_0^1 \varphi + \cancel{x \varphi(x)} \Big|_0^1 - \int_{-1}^0 \varphi + \cancel{x \varphi(x)} \Big|_{-1}^0$$

$$= \int_0^1 \varphi - \int_{-1}^0 \varphi = \int_{-1}^1 \varphi(x) \cdot \underset{\substack{= \\ \begin{cases} +1 & x \in (0, 1) \\ -1 & x \in (-1, 0) \end{cases}}}{\text{sign}(x)} dx$$

$$= \langle T_{\text{sign}(x)}, \varphi \rangle$$

$$\forall \varphi \in \mathcal{D}(-1, 1)$$

$$\boxed{(|x|)' = \text{sign } x \quad \text{in } \mathcal{D}'(-1, 1)}$$



$$3) \quad u(x) = \operatorname{sign}(x) \quad x \in L^1(-1, 1)$$

$$\langle u', \varphi \rangle = - \langle u, \varphi' \rangle \quad \forall \varphi \in \mathcal{D}(-1, 1)$$

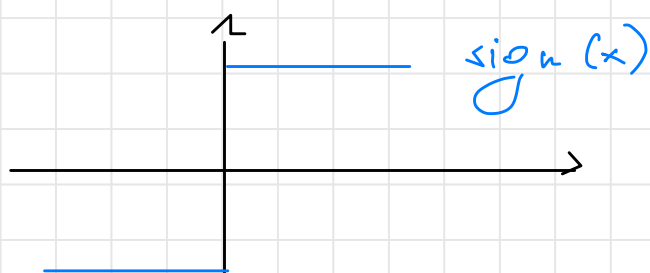
$$= - \int_{-1}^1 \operatorname{sign}(x) \cdot \varphi'$$

$$= - \int_0^1 \varphi' + \int_{-1}^0 \varphi'$$

$$= - [\cancel{\varphi(1)} - \varphi(0)] + [\varphi(0) - \cancel{\varphi(-1)}]$$

$$= 2 \varphi(0) = 2 \langle \delta_0, \varphi \rangle$$

$$u' = 2 \delta_0.$$



$$4) \quad T = \delta_0 \quad T' = ?$$

$$\langle (\delta_0)', \varphi \rangle \stackrel{\substack{= \\ \uparrow \\ \text{per def}}}{=} - \langle \delta_0, \varphi' \rangle = - \varphi'(0)$$

Espazi di Sobolev

$$\begin{cases} -\Delta u = f & \text{eq. Poisson in } \Omega \\ u = 0 & \text{in } \partial\Omega \end{cases}$$



$$\Delta u = u_{x_1 x_1} + u_{x_2 x_2}$$

Def. $\Omega = (a, b) \subseteq \mathbb{R}$, $p \in [1, +\infty]$

$$W^{1,p}(\Omega) := \left\{ u \in L^p(\Omega) : u' \in L^p(\Omega) \right\}$$

↑
derivata distribuzionale

$$u \in L^p(\Omega) \subseteq L^1_{loc}(\Omega) \longrightarrow T_u \in \mathcal{D}'(\Omega)$$

$$(T_u)' \in \mathcal{D}'(\Omega) \quad (T_u)' = T_v \quad \text{con } v \in L^p(\Omega)$$

$$\langle (T_u)', \varphi \rangle = \langle T_v, \varphi \rangle \quad \forall \varphi \in \mathcal{D}(\Omega)$$

$$= \langle T_u'', \varphi \rangle$$

$$= \int_{\Omega} u \varphi'$$

$$\int_{\Omega} v \varphi$$

↑
 u'

Exempi $\Omega = (1, 1)$

• $u \in C_0^1(\Omega) \Rightarrow u \in W^{1,p}(\Omega) \quad \forall p \in [1, +\infty]$

$$\begin{cases} p \in [1, +\infty) & u, u' \in L^p(\Omega) & \int_{\Omega} |u|^p < +\infty, \int_{\Omega} |u'|^p < +\infty \\ p = +\infty & u, u' \in L^\infty(\Omega) & \operatorname{ess\,sup}_{\Omega} |u| < +\infty, \operatorname{ess\,sup}_{\Omega} |u'| < +\infty \end{cases}$$

• $u(x) = |x| \quad u \in W^{1,p}(\Omega) \quad \forall p \in [1, +\infty]$
 $p = 2 \quad u, u' \in L^2(\Omega) \quad u'(x) = \operatorname{sign}(x)$

$$\begin{cases} \int_{\Omega} |x|^2 < +\infty \\ \int_{\Omega} |\operatorname{sign} x|^2 < +\infty \end{cases}$$

• $u(x) = \operatorname{sign}(x) \quad u \notin W^{1,2}(\Omega)$

$$\int_{\Omega} |\operatorname{sign}(x)|^2 < +\infty$$

$u'(x) = 2\delta_0 \in \mathcal{D}'(\Omega) \quad \underline{\text{MA}} \quad u' \notin L^2(\Omega)$

Def. $\Omega = (a, b) \subseteq \mathbb{R}$ $p \in [1, +\infty]$

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) : u' \in L^p(\Omega), u'' \in L^p(\Omega), \dots, u^{(k)} \in L^p(\Omega)\}$$

Def. Ω aperto $\subseteq \mathbb{R}^n$ $p \in [1, +\infty]$

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) : \partial^\alpha u \in L^p(\Omega) \forall \alpha \text{ multiindice con } |\alpha| \leq k\}$$

Caso particolare $p=2$

$$W^{1,2}(\Omega) = H^1(\Omega) = \{u \in L^2(\Omega) : u, u_{x_i} \in L^2(\Omega)\}$$

$$W^{k,2}(\Omega) = H^k(\Omega) = \{u \in L^2(\Omega) : \partial^\alpha u \in L^2(\Omega) \forall |\alpha| \leq k\}$$

Qns. $W^{k,p}(\Omega)$ sono spazi vettoriali

Esempio $k=1$

$$(i) \quad u, v \in W^{1,p}(\Omega) \Rightarrow u+v \in W^{1,p}(\Omega)$$

$$(ii) \quad u \in W^{1,p}(\Omega), \lambda \in \mathbb{R} \Rightarrow \lambda u \in W^{1,p}(\Omega)$$

$$\begin{aligned}
 (i) \quad u+v &\in L^p(\Omega) && \text{vero perché } u, v \in L^p(\Omega) \\
 (u+v)' &\in L^p(\Omega) && \text{vero perché } u', v' \in L^p(\Omega) \Rightarrow \\
 &&& u'+v' \in L^p(\Omega) \\
 &&& (u+v)'
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \lambda u &\in L^p(\Omega) && \text{perché } u \in L^p(\Omega) \\
 (\lambda u)' &\in L^p(\Omega) && \text{perché } u' \in L^p(\Omega) \\
 &&& \lambda \cdot u' \in L^p(\Omega) \\
 &&& (\lambda u)'
 \end{aligned}$$

Def. Norma su $W^{1,p}(\Omega)$

$$\|u\|_{1,p} := \begin{cases} \|u\|_p + \|u'\|_p & n=1 \\ \|u\|_p + \sum_{i=1}^n \|u_{x_i}\|_p & n>1 \end{cases}$$

Def. Norme su $W^{k,p}(\Omega)$

$$\|u\|_{k,p} := \begin{cases} \|u\|_p + \|u'\|_p + \dots + \|u^{(k)}\|_p & n=1 \\ \sum_{|\alpha| \leq k} \|D^\alpha u\|_p & n>1 \end{cases}$$

Oss. $u_h \longrightarrow u$ in $W^{1,p}(\Omega)$

$$\text{se } \|u_h - u\|_{1,p} \longrightarrow 0$$

$$\|u_h - u\|_p + \|u'_h - u'\|_p$$

ovvero

$$\begin{cases} u_h \longrightarrow u & \text{in } L^p(\Omega) \\ u'_h \longrightarrow u' & \text{in } L^p(\Omega) \end{cases}$$

Def.

$$\|u\|_{1,p} = \left(\int_{\Omega} |u|^p \right)^{1/p} + \left(\int_{\Omega} |u'|^p \right)^{1/p} \quad \|u\|_p + \|u'\|_p$$

$$\|u\|_{1,p} = \left(\int_{\Omega} |u|^p + \int_{\Omega} |u'|^p \right)^{1/p} \quad (\|u\|_p^p + \|u'\|_p^p)^{1/p}$$

sono norme equivalenti su $W^{1,p}(\Omega)$

Teorema (di completezza)

Per ogni $p \in [1, +\infty]$, e per ogni $K \in \mathbb{N}$

$W^{K,p}(\Omega)$ sono spazi di Banach

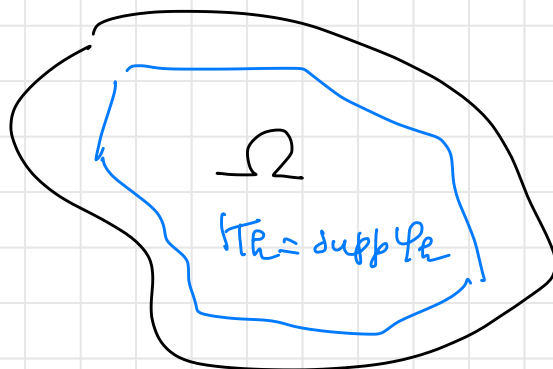
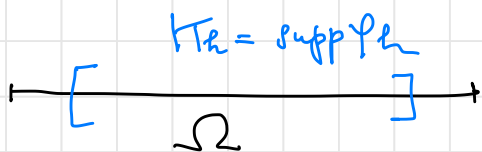
Def. $W_0^{1,p}(\Omega)$ "sottospazio di $W^{1,p}(\Omega)$ delle funzioni
nulle al bordo"

è definito come la chiusura di $C_0^\infty(\Omega)$ in $W^{1,p}(\Omega)$

$$W_0^{1,p}(\Omega) := \overline{C_0^\infty(\Omega)}^{1,p}$$

ovvero

$$u \in W_0^{1,p}(\Omega) \text{ se } \exists \varphi_h \in C_0^\infty(\Omega) : u = \lim_n \varphi_h \\ \| \varphi_h - u \|_{1,p} \longrightarrow 0$$



Dom. Se $u \in C^0(\overline{\Omega}) \cap W^{1,p}(\Omega)$

$$u = 0 \text{ su } \partial\Omega \iff u \in W_0^{1,p}(\Omega)$$

Teorema (disuguaglianza di Poincaré in $W_0^{1,p}(\Omega)$)

Sia Ω aperto **limitato** di $\mathbb{R}^n$

Allora esiste una costante $C_p = C_p(\Omega)$ tale che

$$\|u\|_p \leq C_p(\Omega) \|\nabla u\|_p \quad \forall u \in W_0^{1,p}(\Omega)$$

$$\sum_{i=1}^n \left\| \frac{\partial u}{\partial x_i} \right\|_p$$

Quindi: Su $W_0^{1,p}(\Omega)$

$$\left\{ \begin{array}{l} \|u\|_{1,p} = \|u\|_p + \|\nabla u\|_p \\ \|\nabla u\|_p \end{array} \right\} \text{ sono norme equivalenti.}$$

perché: $\|\nabla u\|_p \leq 1 \cdot (\|u\|_p + \|\nabla u\|_p)$

$$\begin{aligned} (\|u\|_p + \|\nabla u\|_p) &\leq C_p(\Omega) \|\nabla u\|_p + \|\nabla u\|_p \\ &= (C_p(\Omega) + 1) \|\nabla u\|_p. \end{aligned}$$

Es. FALSA su $W_0^{1,p}(\Omega)$

$$\Omega = (-1, 1) \quad u = \lambda \neq 0$$

Es. $n=1$ $u \in C^1(\overline{\Omega})$ $\Omega = (0, 1)$ $p=2$
se $u(0) = u(1) = 0$

$$u(x) = \cancel{u(0)} + \int_0^x u'(t) dt$$

$$|u(x)| \leq \int_0^x |u'(t)| dt \leq \int_0^1 |u'(t)| dt \stackrel{\text{Hölder}}{\leq}$$

$$\leq \left(\int_0^1 |u'(t)|^2 \right)^{1/2} = \|u'\|_{L^2(0,1)}$$

Ho ottenuto : $|u(x)|^2 \leq \|u'\|_2^2 \quad \forall x \in (0,1).$

Integro su $(0,1)$: $\int_0^1 |u(x)|^2 \leq \int_0^1 |u'(x)|^2$

$$\|u\|_2^2 \leq \|u'\|_2^2$$

Valo la Poincaré con $C_2(\Omega) = 1.$

Spazi di Hilbert

Def. Sia H uno spazio vettoriale

Un **PRODOTTO SCALARE** su H è un'applicazione

$(\cdot, \cdot) : H \times H \longrightarrow \mathbb{R}$ tale che:

(i) $(x, x) \geq 0 \quad \forall x \in H$, con $(x, x) = 0 \iff x = 0$
POSITIVITA'

(ii) $(x, y) = (y, x) \quad \forall x, y \in H$ SIMMETRIA

(iii) $(\alpha_1 x_1 + \alpha_2 x_2, y) = \alpha_1 (x_1, y) + \alpha_2 (x_2, y)$
 $\forall x_1, x_2, y \in H, \quad \forall \alpha_1, \alpha_2 \in \mathbb{R}$ BILINEARITA'

Dim. Segue da (ii) + (iii) :

$$(i) \quad \xrightarrow{\quad} \quad (x, \alpha_1 y_1 + \alpha_2 y_2) = \alpha_1 (x, y_1) + \alpha_2 (x, y_2)$$

$$(ii) \quad \xrightarrow{\quad} \quad (\alpha_1 y_1 + \alpha_2 y_2, x)$$

$$(ii) \quad \xrightarrow{\quad} \quad \alpha_1 (y_1, x) + \alpha_2 (y_2, x)$$

$$\alpha_1 (x, y_1) + \alpha_2 (x, y_2)$$

Def. $\|x\| := \sqrt{(x, x)}$ NORMA INDOTTA DA $(,)$

• $\|x\| \geq 0$ con $= \Leftrightarrow x=0$ ✓

• $\|\lambda x\| = |\lambda| \|x\|$

$$\|\lambda x\| = \sqrt{(\lambda x, \lambda x)} = \sqrt{\lambda^2 (x, x)} = |\lambda| \sqrt{(x, x)} = |\lambda| \|x\| \quad \checkmark$$

• $\|x + y\| \leq \|x\| + \|y\|$

↓
segue dalla diseg + CS

Disuguaglianza di Cauchy-Schwartz

Sia H sp. vettoriale con $(,)$. Allora

$$|(x, y)| \leq \|x\| \cdot \|y\| \quad \forall x, y \in H$$

con $= \Leftrightarrow x = \lambda y$ con $\lambda \in \mathbb{R}$

Dim. $\forall t \in \mathbb{R}$

$$0 \leq \underbrace{(x - ty, x - ty)}_{(i)} = \|x\|^2 - 2t(x, y) + t^2 \|y\|^2$$

$$\Rightarrow \Delta \leq 0 \quad \text{ovvero} \quad 4(x, y)^2 - 4\|x\|^2 \|y\|^2 \leq 0$$

$$\text{cioè} \quad (x, y)^2 \leq \|x\|^2 \|y\|^2$$

Caso uguaglianza

$$\text{Se vale} \quad |(x, y)| = \|x\| \cdot \|y\|, \quad \Delta = 0$$

$$\Rightarrow \exists \lambda \in \mathbb{R} : (x - \lambda y, x - \lambda y) = 0$$

$$\Rightarrow \text{per (i), } x - \lambda y = 0 \quad x = \lambda y$$

Viceversa

$$\text{Se } x = \lambda y : \begin{cases} |(x, y)| = |(\lambda y, y)| = |\lambda| \|y\|^2 \\ \|x\| \cdot \|y\| = \|\lambda y\| \cdot \|y\| = |\lambda| \|y\|^2 \end{cases}$$

Dim. triangolare

$$\begin{aligned} \|x + y\| &= \sqrt{(x+y, x+y)} = \sqrt{\|x\|^2 + \|y\|^2 + 2(x, y)} \\ \text{CS } \wedge \quad \|x\| + \|y\| &= \sqrt{(\|x\| + \|y\|)^2} \\ &= \sqrt{\|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\|} \end{aligned}$$

□

Esempi

$$\begin{aligned} \bullet \quad H = \mathbb{R}^m \quad (x, y) &= x \cdot y = \sum_{i=1}^m x_i y_i \\ \|x\|_2 &= \sqrt{(x, x)} = \sqrt{\sum_{i=1}^m x_i^2} \end{aligned}$$

$$\begin{aligned} \bullet \quad H = \ell^2 \quad (x, y) &= \sum_{i \in \mathbb{N}} x_i y_i \\ \|x\|_2 &= \sqrt{(x, x)} = \sqrt{\sum_{i \in \mathbb{N}} x_i^2} \end{aligned}$$

- $H = L^2(\Omega)$ $(f, g) = \int_{\Omega} f g$ (ben definito per Hölder)

$$\|f\|_2 = \left(\int_{\Omega} f^2 \right)^{1/2}$$

- $H = H^1(\Omega)$
 $= W^{1,2}(\Omega)$

$$(f, g) = \int_{\Omega} f g + \int_{\Omega} \nabla f \cdot \nabla g$$

pr. scalare in $\mathbb{R}^n$

$$\|f\|_{1,2} = \left(\int_{\Omega} f^2 + \int_{\Omega} |\nabla f|^2 \right)^{1/2}$$

Osservazione:

$$\|\cdot\|_p \text{ in } \mathbb{R}^n$$

$$\|\cdot\|_p \text{ in } \mathbb{R}^p$$

$$\|\cdot\|_p \text{ in } L^p(\Omega)$$

$$\|\cdot\|_p \text{ in } W^{k,p}(\Omega)$$

con $p \neq 2$

non provengono da
un prodotto scalare

Legge del parallelogramma

Sia H sp. vettoriale con prodotto scalare e

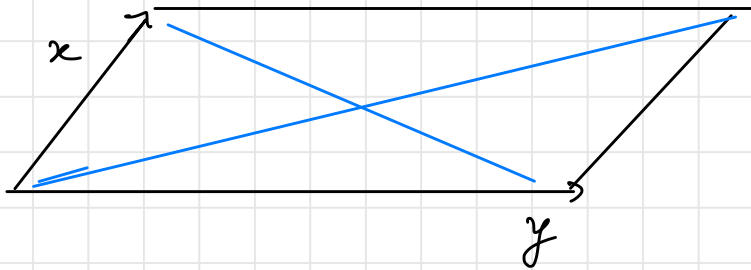
e sia $\| \cdot \|$ la norma indotta dal prodotto scalare

Allora

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

$$\forall x, y \in H.$$

In $\mathbb{R}^2$



dim.

$$\begin{aligned} \bullet \quad & \|x+y\|^2 + \|x-y\|^2 = (x+y, x+y) + (x-y, x-y) \\ &= \|x\|^2 + 2 \cancel{(x,y)} + \|y\|^2 \\ &+ \|x\|^2 - 2 \cancel{(x,y)} + \|y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \square \end{aligned}$$